

Integrated GPS/Inertial Fault Detection Availability

MATS BRENNER

Honeywell, Minneapolis, Minnesota

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ABSTRACT

Different techniques have been proposed for combining GPS and inertial sensor information, and recent published findings seem to indicate that primary-means integrity availability is achievable with standard 2 nmi/h (95 percent) inertial sensor performance. This paper investigates and quantifies the different inertial effects that contribute to enhanced integrity of the integrated GPS/inertial system, such as coasting or Schuler feedback. A Kalman filter-based integration scheme that preserves the integrity information in an optimal fashion is presented and used to quantify integrity performance. This paper also proposes an approximate model, incorporating the main inertial effects contributing to integrity, that can be used to calculate the achievable horizontal protection level (HPL) at any geographical location and time. This model is used to estimate the availability of fault detection for an integrated GPS/inertial system. In addition, the paper compares the availability of fault detection for the GPS/inertial system with that for other augmentations to provide trade-off information.

INTRODUCTION

GPS systems possess a high availability of accurate horizontal positioning, but fault detection and exclusion (FDE) techniques based on satellite redundancy do not provide availability sufficient for primary-means navigation in the terminal and nonprecision approach phases of flight. Different types of augmentations, such as WAAS, Loran, baroaiding, atomic clock, and inertial aiding, are under consideration for achieving primary-means integrity availability. The availability of FDE will depend on the geometry of the satellites used for positioning. The availability of satellites in good geometry is in turn a function of the status of the GPS constellation (e.g., failures and maintenance). Methods for availability calculation have been developed and adopted by RTCA Special Committee (SC)-159, and the availability of FDE over a specified region can be calculated for most augmentations. However, corresponding estimates (conforming with RTCA SC-159 guidelines) of the FDE availability provided by GPS/inertial integration techniques have not yet been published. Integrity requirements and associated test procedures that meet future needs have been developed for autonomous GPS receivers. However, because of the complexity of hybrid GPS/inertial systems, clear integrity requirements and appropriate test procedures have not yet been developed for these systems.

As affordable computing capacity in embedded applications has become increasingly available, a renewed interest in integrated GPS/inertial systems for meeting the anticipated RTCA/FAA requirements for primary-means GPS-based navigation has emerged. This paper describes one possible implementation and demonstrates how the availability of integrity monitoring, which is a major certification issue, can be determined. Test procedures that would be applicable to the proposed implementation are also included.

FDE REQUIREMENTS AND ASSUMPTIONS

The receiver autonomous integrity monitoring (RAIM) requirements published in RTCA DO-208 are used throughout this paper. The false detection rate (p_{fd}) is set to 0.002/h, and the missed detection probability (p_{md}) is 0.001. Alert limits are 0.3 nmi for nonprecision approach, 1 nmi for terminal, and 2 nmi for en route, and (deviating from DO-208) 5–10 nmi for oceanic phases of flight.

A failure in a satellite that causes the horizontal error to exceed the alert limit for the phase of flight at hand must be announced to the pilot. The probability that the position error exceeds the alert limit before annunciation occurs must be smaller than or equal to p_{md} . Although multiple satellite failures most likely will be announced, praxis mandates that only single satellite failures need be considered when the integrity performance is evaluated. The false detection rate is internal to the algorithm, and only false detections indicating failures will cause false alerts.

No assumptions can be made regarding the dynamics of the satellite failure, such as whether it is a ramp, oscillation, or random walk. This ground rule is crucial in the described implementation.

The dominant measurement error is the selective availability (SA) noise. For test purposes, the total system error is modeled as a second-order Markov process (see [1]) and a bias. The second-order process has the following characteristics. The time constant is 118 s, the damping factor is $1/\sqrt{2}$, and the pseudorange-rate sigma σ_{pr} is $0.1217 \times \sigma_{pr}$. The pseudorange sigma σ_{pr} varies depending on the bias such that $\sqrt{\sigma_{pr}^2 + \sigma_{bias}^2} \leq 33.3$ m. The ratio of bias to dither is not known. However, SA bias components are rarely observed. The analysis in this paper is directed primarily toward the impact of dithered SA, but bias components are also briefly discussed. This SA model has been established for certification testing by RTCA. The details of the true SA model are not known.

The satellite constellation used is the optimized 24 satellite constellation. The constellation weights, taken from [4], and the representative average cases (see [7]) used are shown in Table 1. The weights for 21 and fewer satellites are accumulated in the 21 satellite case.

Table 1—Constellation Weights and Average Cases

Number of Satellites	Weight	Excluded Satellites
24	0.703	N/A
23	0.227	A3
22	0.055	A1,F3
21	0.015	A2,E3,F2

An essential error component in a hybrid system is the local variations of gravity. Deviations in the vertical direction (Δg_z) from the nominal vertical g are characterized as gravity anomalies and measured in parts per million (ppm). Deviations in the horizontal directions ($\Delta v_x, \Delta v_y$) are characterized as gravity deflections and measured in arcseconds. The numbers used in this paper are shown in Table 2.

A SOLUTION SEPARATION-BASED RAIM ALGORITHM

Most snapshot RAIM algorithms are derivable from parity space theory. In such algorithms, the monitoring takes place either in the measurement domain or in the parity space. The impact of a failure is calculated by projecting the estimated worst-case offset into the horizontal plane. It is demonstrated here that one can create a RAIM algorithm based purely on solution separation in the horizontal plane. The approach presented is somewhat different from that presented in [4], and incorporates derivations of threshold setting and calculation of horizontal protection level. The protection level derived this way has been shown to be almost identical to that calculated by the Honeywell parity space algorithm, which is a refined version of the algorithm presented in [5]. The algorithm presented in [5] has been further refined to account properly for the SA noise in all involved satellites, not just the satellite in which a possible failure is considered.

The solution separation algorithm proposed in this paper is derived as follows. Let $\mathbf{S}_0$ be the least-squares solution matrix for a full set of N measurements. Let $\mathbf{S}_n$ be the solution matrix for the subset where satellite n is excluded. Construct $\mathbf{S}_n$ so that it operates on the full set of measurements by extending the $4 \times (N - 1)$ subsolution matrix with an n^{th} zeroed column.

The full solution is

$$\Delta \mathbf{x}_0 = \mathbf{S}_0 \Delta \mathbf{p} \quad (1)$$

The subsolutions are

$$\Delta \mathbf{x}_n = \mathbf{S}_n \Delta \mathbf{p}, \quad (n = 1, N) \quad (2)$$

$\Delta \mathbf{x}$ denotes the position correction to the initial estimate in a local geographic north-east-down (NED) coordinate system. The fourth component is the receiver time correction times the speed of light. $\Delta \mathbf{p}$ is the N dimensional measurement vector relative to the initial estimate. The discriminator d_n is defined as

$$d_n = \sqrt{(\Delta x_0(1) - \Delta x_n(1))^2 + (\Delta x_0(2) - \Delta x_n(2))^2} \quad (3)$$

The impact of the SA noise in the full solution is

$$\delta \mathbf{x}_0 = \Delta \mathbf{x}_0 - \Delta \mathbf{x} = \mathbf{S}_0 \mathbf{w} \quad (4)$$

Table 2—Gravity Anomalies and Deflections

Gravity Component	1-sigma	Correlation Distance
Anomaly	25 ppm	20 nmi
Deflections	5 arcsec	20 nmi

The SA noise impact in the subsolutions is

$$\delta \mathbf{x}_n = \Delta \mathbf{x}_n - \Delta \mathbf{x} = \mathbf{S}_n \mathbf{w}, \quad (n = 1, N) \quad (5)$$

$\delta \mathbf{x}$ denotes the position error relative to the actual position offset $\Delta \mathbf{x}$. $\mathbf{w}$ is the N dimensional measurement noise vector. $E[\mathbf{w} \mathbf{w}^T] = \mathbf{I} \sigma_{pr}^2$.

The covariance matrix describing the statistics of the separation between the full filter and the subfilter is defined as

$$\mathbf{dP}_n = E[(\Delta \mathbf{x}_0 - \Delta \mathbf{x}_n)(\Delta \mathbf{x}_0 - \Delta \mathbf{x}_n)^T] \quad (6)$$

and can be calculated as follows

$$\mathbf{dP}_n = (\mathbf{S}_0 - \mathbf{S}_n) \sigma_{pr}^2 (\mathbf{S}_0 - \mathbf{S}_n)^T \quad (7)$$

Because of the directional character of the separation distribution, one eigenvalue will dominate, and a stringent upper bound of the discriminator variance is the maximum eigenvalue λ^{dp_n} of the horizontal position separation distribution. This eigenvalue can be calculated from the covariance matrix elements $dp_n(1,1)$, $dp_n(1,2)$, and $dp_n(2,2)$.

Considering both sides of the distribution and the fact that any of the N filters may cause a false detection, the threshold D_n (Figure 1) is calculated as

$$D_n = \sqrt{\lambda^{dp_n}} Q^{-1} \left(\frac{p_{fd}}{2N} \right) \quad (8)$$

where Q^{-1} is the inverse of

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^\infty e^{-\frac{t^2}{2}} dt \quad (9)$$

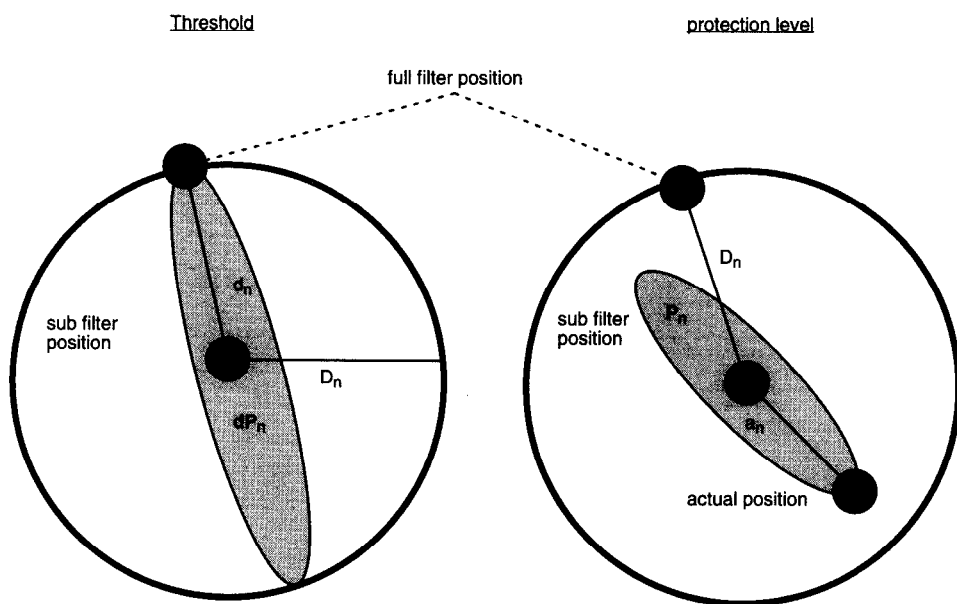


Fig. 1—Horizontal Distributions

The covariance matrices P_n describing the error in the subfilters, defined as

$$P_n = E[\delta \mathbf{x}_n \delta \mathbf{x}_n^T], \quad (n = 1, N) \quad (10)$$

are calculated as

$$P_n = S_n \sigma_{pr}^2 S_n^T, \quad (n = 1, N) \quad (11)$$

At detection (see Figure 1), the separation between the full filter and the subfilter is D_n . The distribution of the actual horizontal position versus the subfilter position is described by P_n . A stringent upper bound of the error variance along the direction of the separation D_n is given by the maximum eigenvalue λ^{P_n} . The error bound a_n corresponding to the missed detection probability p_{md} is

$$a_n = \sqrt{\lambda^{P_n}} Q^{-1}(p_{md}) \quad (12)$$

a_n and D may act in different directions (see Figure 1), but an upper limit on the full filter position error at detection is given by $D + a_n$.

The horizontal protection level is then

$$HPL_{cov} = \max\{D_n + a_n\} \text{ over } n = 1, N \quad (13)$$

The horizontal protection level HPL_{par} calculated using the current Honeywell parity space algorithm and the covariance-based horizontal protection level HPL_{cov} are both plotted in Figure 2.

It is an important finding that these HPLs, although calculated by fundamentally different algorithms employing seemingly different upper-bound approximations, are nearly identical.

GENERALIZATION TO BANK OF KALMAN FILTERS

When integrating inertial systems with GPS, the complexity of the inertial dynamics suggests Kalman filter-based implementations. The major difference

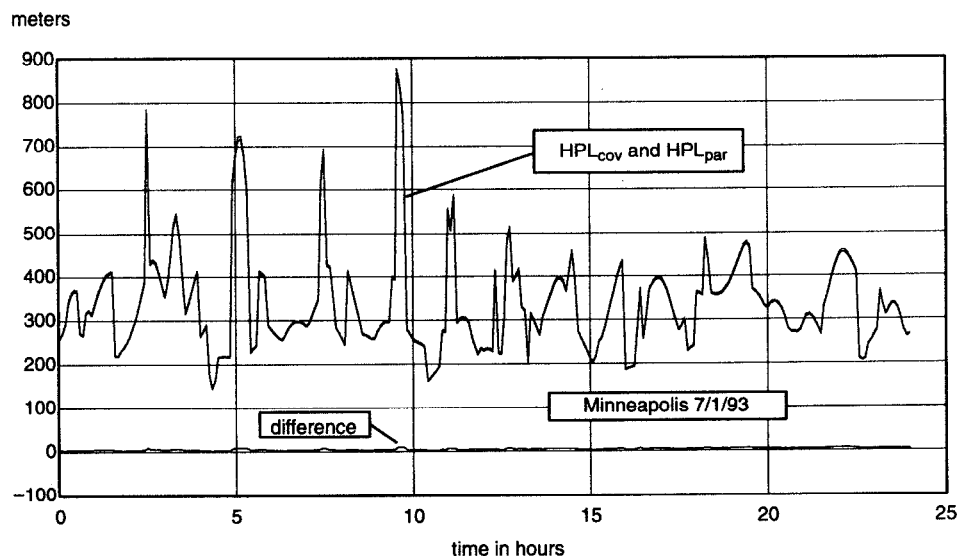


Fig. 2—Protection Level Comparison

between the covariance-based RAIM algorithm and the parity space algorithm is that it is possible to generalize the former to Kalman filter-based GPS implementations. As pointed out in [6, 7], the integrity assurance will require a separate sub-Kalman filter for each satellite. This paper proposes that a subfilter and a covariance propagator be assigned to each tracked satellite in order to rigorously incorporate the basic fault detection requirements. The subfilter is a straightforward Kalman filter producing a position solution with one satellite excluded. The covariance propagator calculates the statistics of the separations between the full filter and the subfilters.

The rationale for this is as follows. Assume that the state vector (see Appendix A) for the full solution is $\Delta \mathbf{x}_0(t)$, and the corresponding covariance matrix is $\mathbf{P}_0(t)$. Usually, the dimension of the state vector will be at least 18, so $\mathbf{P}_0$ is at a minimum an 18×18 matrix. The dependence of time is included to emphasize that the positioning filtering occurs over time, as opposed to the least-squares snapshot solution discussed in the previous section. N sub-Kalman filters with covariance matrices $\mathbf{P}_n(t)$ excluding satellite n and providing solutions $\Delta \mathbf{x}_n(t)$ are generated. The N solution separations $\mathbf{d}\mathbf{x}_n(t)$ are calculated:

$$\mathbf{d}\mathbf{x}_n(t) = \Delta \mathbf{x}_0(t) - \Delta \mathbf{x}_n(t) \quad (14)$$

The covariance propagator calculates the covariance matrix $\mathbf{d}\mathbf{P}_n(t)$ describing the statistics of the separations between the full filter and the subfilters.

$$\mathbf{d}\mathbf{P}_n(t) = E[(\Delta \mathbf{x}_0(t) - \Delta \mathbf{x}_n(t)) (\Delta \mathbf{x}_0(t) - \Delta \mathbf{x}_n(t))^T] \quad (15)$$

For each satellite measurement that is processed, this calculation uses the Kalman gain vector $\mathbf{g}_n$ from the n^{th} subfilter and the gain vector $\mathbf{g}_0$ from the full filter to create the $2N \times 2N$ covariance matrix $\mathbf{P}_n^{\text{dual}}(t)$. This is further described in Appendix B. The covariance matrix $\mathbf{P}_n^{\text{dual}}(t)$ can be partitioned as

$$\mathbf{P}_n^{\text{dual}}(t) = \begin{bmatrix} \mathbf{P}_0(t) & \text{cross corr} \\ \text{cross corr} & \mathbf{d}\mathbf{P}_n(t) \end{bmatrix} \quad (16)$$

The separation statistics are described by $\mathbf{d}\mathbf{P}_n(t)$.

THRESHOLD SETTING AND HORIZONTAL PROTECTION LEVEL

The matrix elements of interest are the elements representing the horizontal position separation and error statistics. The error state vector (see Appendix A) will carry these in positions 7 and 8. Using the results from the previous section, the two maximum eigenvalues are determined. $\lambda \mathbf{d}\mathbf{P}_n(t)$ is calculated from the covariance matrix elements $\mathbf{d}p_n(7,7)$, $\mathbf{d}p_n(7,8)$, $\mathbf{d}p_n(8,8)$ describing the statistics of the filter separations in the horizontal plane. $\lambda^{P_n}(t)$ is calculated from the covariance matrix elements $p_n(7,7)$, $p_n(7,8)$, $p_n(8,8)$ describing the statistics of the subfilter position error.

For each subfilter, a discriminator is defined as

$$d_n = \sqrt{(\Delta \mathbf{x}_0(7) - \Delta \mathbf{x}_n(7))^2 + (\Delta \mathbf{x}_0(8) - \Delta \mathbf{x}_n(8))^2} \quad (17)$$

A variable threshold $D_n(t)$ is set based on the current subfilter versus full filter separation statistics represented by the upper bound variance $\lambda \mathbf{d}\mathbf{P}_n(t)$ in full correspondence with equation (8).

$$D_n(t) = \sqrt{\lambda^{dP_n}(t)} Q^{-1} \left(\frac{p_{fd}}{2N} \right) \quad (18)$$

Likewise, an upper bound on the subfilter error, commensurate with the missed detection probability, is calculated as

$$a_n(t) = \sqrt{\lambda^{p_n}(t)} Q^{-1}(p_{md}) \quad (19)$$

By analogy with equation (13), it now follows that the horizontal protection level for the GPS/inertial hybrid system is

$$HPL_{hyb} = \max\{D_n(t) + a_n(t)\} \text{ over } n = 1, N \quad (20)$$

The simulated HPL_{hyb} for a 2 nmi/h inertial system is plotted in Figure 3, together with the snapshot-based HPL_{par} . The time step is 120 s between each set of satellite measurement updates. The HPL_{hyb} grows between the Kalman filter measurement updates as a result of the uncertainty in the estimated error states. This causes a sawtooth appearance.

Just after 50 min (4:50), a 0.3 nmi integrity hole occurs, and the snapshot RAIM HPL_{par} goes up to 560 m. The subfilters (in particular the subfilter representing the worst-case satellite impact as shown in Figure 3) will be unaffected by any failure in this excluded satellite, and will make use of the optimal estimates of the inertial error states, in combination with additional measurements from satellites now in a less favorable geometry, to carry the truth reference through this integrity hole. In this particular case, the hybrid protec-

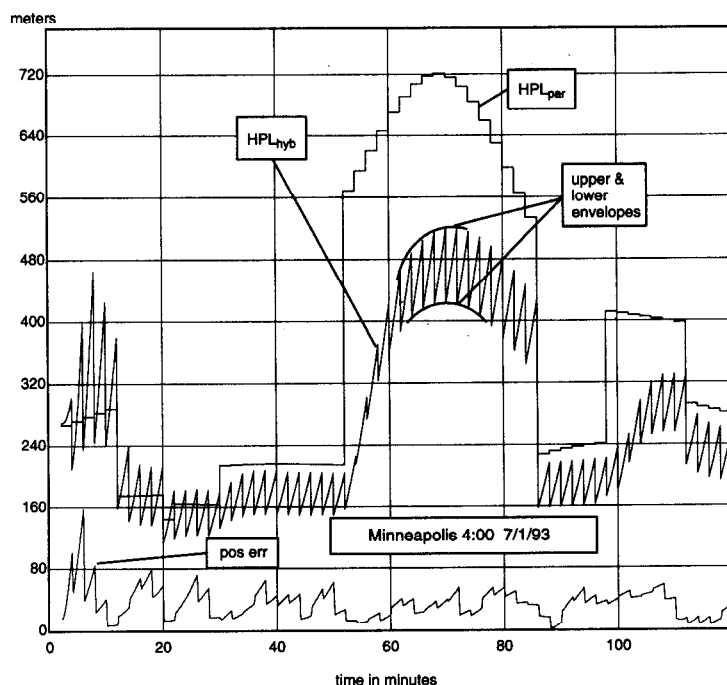


Fig. 3—Coasting Through an Integrity Hole

tion level will stay below the 550 m required for nonprecision approach. The method suggested in this paper for determining the availability of the hybrid integrity is (1) to demonstrate that HPL_{hyb} (lower envelope) is always less than HPL_{par} , and (2) (as demonstrated in Figure 3) to determine for each RAIM integrity hole whether the hybrid integrity will meet the alert limit for the particular phase of flight under consideration.

The growth in HPL_{hyb} between Kalman corrections causes a variation in the corrected and uncorrected HPL of up to 80 m (width of sawtooth pattern). The reason measurements are taken 2 min apart is that such SA samples can be assumed to be uncorrelated. The 2 min sample time is also related to the capacity of the on-board hybrid computer and results in an acceptable CPU load. Theoretically, though, since GPS measurements are available every second, nothing would prevent a series of independent banks of filters, shifted in time, all using a 2 min time step, that provided fully Kalman-corrected error states every 5 s. Availability results will therefore be provided both for the upper envelope and for the lower envelope. If further SA modeling details in addition to the 2 min decorrelation assumption, such as the second-order characteristics of the dither, are pursued, the subfilters in this approach can be extended with additional SA states, which would allow the update time to be shortened to seconds. However, to ensure compliance with available SA information, the results in this investigation are based on the the 2 min SA decorrelation only; that is, no further assumptions regarding the dynamics of the SA dither are used.

The simulation program that produced the result in Figure 3 models the airplane dynamics and runs with a 0.1 s time step. This program contains the bank of the hybrid subfilters and covariance propagators used for detection and HPL calculations. The number of integrity holes with a 5 deg mask angle ranges from 5,349 for 24 satellites to 23,687 for 21 satellites. Because of time constraints, it is currently not possible to run this high-fidelity simulation for all integrity holes. One major result of this paper is the characterization of the basic effects responsible for the hybrid integrity and the creation of a simplified model that approximately predicts the coasting performance demonstrated in Figure 3. This simplified model will use a 2 min time step and will run rapidly enough to be used for the availability determination.

Typical sensor errors for a 2 nmi/h inertial system are summarized in Table 3.

SCHULER FEEDBACK, GRAVITY FEEDBACK, AND COASTING

To create a simplified model, the basic effects that provide the hybrid integrity enhancements were determined. An extensive simulation effort has rank-ordered the effects as shown in Table 4.

Table 3—Typical Inertial Sensor Parameters

Gyro random walk	0.003°/h ^{1/2}
Gyro bias	0.008°/h
Gyro scale factor	5 ppm
Gyro misalignment	17 arcsec
Acceleration bias	80 μ g
Acceleration scale factor	300 ppm
Acceleration misalignment	17 arcsec

Table 4—Ordered Inertial Effects

1	Inertial coasting
2	Vertical gravity/Schuler feedback coupling
3	Vertical gravity feedback/clock coupling
4	Schuler feedback/clock coupling

Note that the effect responsible for the estimation of SA biases in a Kalman filter using separate bias states for each satellite is due to the satellite motion and is not related to any inertial effect. Such bias estimation requires long-term stability of the SA biases. There is currently no DOD- or FAA-endorsed SA bias specification available.

Inertial Coasting

Inertial coasting is the main contributor to integrity. For any single satellite failure, there is always at least one subfilter that has been calibrated for a long time with measurements that are completely unaffected by this satellite failure. Even if the failure has been in progress over a long time, the subfilter is still providing a true reference for integrity monitoring. Inertial coasting was illustrated in Figure 3.

Schuler Feedback/Receiver Clock Coupling

Any horizontal position error will induce a velocity error, which in turn will reduce the position error and subsequently reverse the sign of the initial error. A clock phase error will have different dynamic characteristics. The hybrid sub-Kalman filters can use such characteristics to reduce the position error over time even if only three satellites in good geometry are available in each subfilter. Given the gyro drift and the unfavorable correlations created during alignment, the protection level that can be obtained based on the Schuler feedback/receiver clock coupling by a 2 nmi/h system using only four satellites is about 56,000 m (30 nmi).

Vertical Gravity Feedback/Clock Coupling

An inertial system incorporates a gravity model. The accuracy of this model is reflected in the gravity anomaly and deflection first-order processes in Table 2. The gravity model includes a height (h) dependence based on the familiar inverse square law. An offset in altitude will cause a residual z-acceleration. The relation between the residual acceleration and the height offset is (see Appendix A):

$$dg = -2/R g(0) dh \tag{21}$$

where R is earth radius, and g(0) is gravity at h = 0. In a stationary application, the height can be measured by processing the z-accelerometer outputs. In a dynamic application, the vertical channel is inherently unstable. If the vertical channel is combined with a receiver clock model using a 0.1 m/s/√s random walk in clock frequency (6-state model), and measurements are performed that observe the sum of vertical position and clock phase error, the different

characteristics of the clock and the gravity feedback allow the Kalman filter to pull in the vertical position and clock phase error to about 2500 m (1σ). This limit depends strongly on the clock performance, but not much on z-accelerometer accuracy in the interval 40–300 μg .

Vertical Gravity/Schuler Feedback Coupling

To demonstrate this coupling effect, the vertical channel using a 40 μg z-accelerometer and the clock (6 states) are combined with the Schuler dynamics of one of the horizontal channels (6 states) in a 12-state Kalman filter. Measurements that are the sum of the vertical position and the clock phase error, combined with measurements that are the sum of the lateral position and the same clock phase error, are performed. The vertical position error will be pulled in to about 300 m (1σ). This result is not sensitive to the clock performance, but is strongly related to the z-accelerometer accuracy. The Kalman filter was able to eliminate the clock phase error because of the radically different dynamics of the vertical and horizontal channels. The hybrid Kalman filter will automatically incorporate this type of information. This means integrity below 1 nmi can be established with only four satellites (in good geometry) in view and no altitude aiding. It can be demonstrated that an accurate z-accelerometer is approximately equivalent to an altitude measurement. The z-accelerometer versus altitude (1σ) accuracy relation is summarized in Table 5.

ALTITUDE AIDING

Further benefits are provided by baro-corrected altitude (50 m, 1σ accuracy) and calibrated pressure altitude (150–250 m, 1σ). The bank of Kalman filters and covariance propagators automatically incorporates pressure altitude aiding. This is required to ensure the stability of the solution if satellites are temporarily lost. Since every subfilter has its own private pressure altitude bias unaffected by the excluded satellite, the integrity of the bias correction is automatically protected. For a 2 nmi/h system, the benefit of altitude aiding (calibrated or baro-corrected) will exceed any of the inertial coupling effects described, except in special circumstances when pressure altitude calibration has not yet been performed, and baro-corrections are not available.

SIMPLIFIED MODEL

Availability is calculated by using a worldwide grid of test points combined with time sampling over 24 h. The grid elements are 180 nmi $\times$ 180 nmi in size, and the time step used is 5 min. The total number of time-space points N_{tot} is 681,120. For each of the four constellations (indexed by $i = 0,3$) in Table 1, the protection level calculated at each time-space point is compared with the alert limit. The number of time-space points where the protection

Table 5—Equivalent Altitude Accuracy

Acceleration	300 μg	100 μg	40 μg	20 μg
Altitude	920 m	420 m	300 m	280 m

level is less than the alert limit is N_i . The weights in Table 1 are denoted w_i ($i = 0, 3$). The availability A is calculated as

$$A = \sum_{i=0}^3 w_i \frac{N_i}{N_{\text{tot}}} \quad (22)$$

The location, time, and duration of all contiguous integrity holes are stored on disc for each of the four constellations. To investigate the availability of augmentations involving calibration such as pressure altitude aiding, atomic clock aiding, and hybrid GPS/inertial filtering, only the holes are run, with allowance for sufficient calibration time before the hole starts. The number of holes is considerable, and a simplified HPL_{hyb} model as described below has been developed for this purpose.

It is a known fact that GPS/inertial Kalman filters can be designed to process either pseudoranges or position information from a GPS receiver. Even if the latter method is not theoretically correct, since position errors in the north and east directions are sometimes correlated, it usually works. In the simplified approach, equations (1) and (2) are used to create the full least-squares solution $\Delta \mathbf{x}_0$ and the sub-least-squares solutions $\Delta \mathbf{x}_n$ (one for each excluded satellite n). The subsolution covariance matrix $\mathbf{P}_n$ (equation (10)) and associated worst-case variance λ^{P_n} , as well as the separation covariance matrix $d\mathbf{P}_n$ (equation (7)) and the associated worst-case variance λ^{dP_n} , are calculated. A bank of 6-state Kalman filters and 6-state covariance propagators is used to model the inertial dynamics in the (horizontal) direction of the worst-case variances, as provided by the least-squares subsolutions and least-squares separations. The 6-state vector is chosen as (see Appendix A)

$$\Delta \mathbf{x}^T = (\psi_y^L, \Delta v_x^L, \Delta r_x^L, \Delta \omega_{\delta y}^B, \Delta a_{\delta x}^B, \Delta v_x) \quad (23)$$

The Kalman filter measurement is defined by

$$\mathbf{h}^T = (0, 0, 1, 0, 0, 0) \quad (24)$$

The covariance matrix generated by the 6-state subfilter is $\mathbf{P}_n^a(k)$. At the beginning of the availability run, $\mathbf{P}_n^a(k)$ is initialized with a 300 m position sigma, a 0.1 m/s velocity sigma, and a 0.1 mrad attitude sigma, and inertial sensor errors (1σ) are initialized according to Table 3. The element $\mathbf{P}_n^a(3,3)$ approximates the worst-case variance λ^{P_n} of the hybrid subfilter. The covariance matrix generated by the covariance propagator is $d\mathbf{P}_n^a(k)$. This matrix is initialized to $\mathbf{0}$, and $d\mathbf{P}_n^a(3,3)$ corresponds to the worst-case variance λ^{dP_n} of the hybrid covariance propagator. The horizontal protection level $\text{HPL}_{\text{hybavl}}$ is calculated (following equations (8)–(13)) as

$$\text{HPL}_{\text{hybavl}} = \max\{D_n + a_n\}, \quad (\text{over } n = 1, N) \quad (25)$$

where

$$D_n = \sqrt{d\mathbf{P}_n^a(3,3)} \mathbf{Q}^{-1} \left(\frac{P_{\text{fd}}}{2N} \right) \quad (26)$$

$$a_n = \sqrt{\mathbf{P}_n^a(3,3)} \mathbf{Q}^{-1}(p_{\text{md}}) \quad (27)$$

The logic used for initializing and releasing these 6-state filters and covariance propagators as satellites are lost and gained is the same as for the complete hybrid integrity filters and propagators (see Appendix B).

The performance of the simplified model is demonstrated in Figure 4 without pressure altitude aiding active. The simplified model accurately reflects the essential dynamics of the hybrid horizontal protection level. This demonstrates that the basic effects involved are understood.

In Figure 5, two satellites are set unhealthy, and pressure altitude aiding is incorporated in the simplified model. In each subfilter, this is done by including an altitude measurement with an associated sigma σ_{altb} that is calibrated to $\sigma_{altb} = \text{VDOP (pure GPS)} \times \sigma_{pr} \times \text{fac}$ when the condition

$$\text{VDOP (pure GPS)} \times \sigma_{pr} \times \text{fac} < \sigma_{altb}$$

is met. σ_{altb} is propagated in compliance with FAA TSO-C129. The factor $\text{fac} (= 4)$, compensating for the nonwhite character of the calibrated bias, is determined by matching $\text{HPL}_{\text{hybavl}}$ with HPL_{hyb} . In this case, the nonprecision approach integrity hole was of a type that cannot be overcome by pressure altitude aiding and coasting in combination. Although the complete model can incorporate bias states, the simplified model has not yet been refined to do so. The availability results presented below assume that the SA noise is dithered.

AVAILABILITY RESULTS

The simplified model has been used to determine the availability of the hybrid integrity. This model is run through each registered integrity hole and

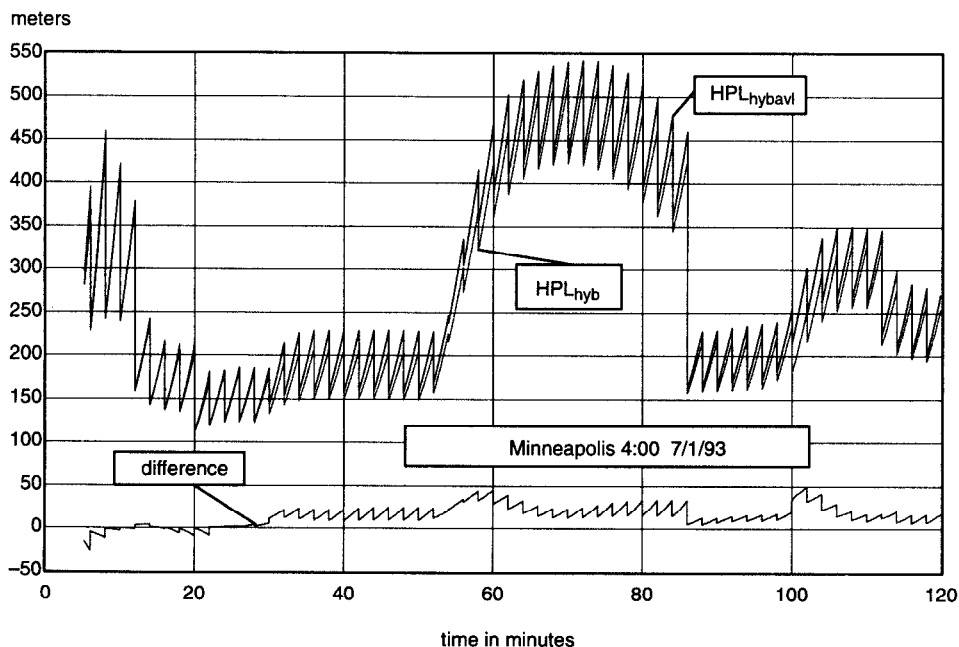


Fig. 4—Performance of Simplified Model

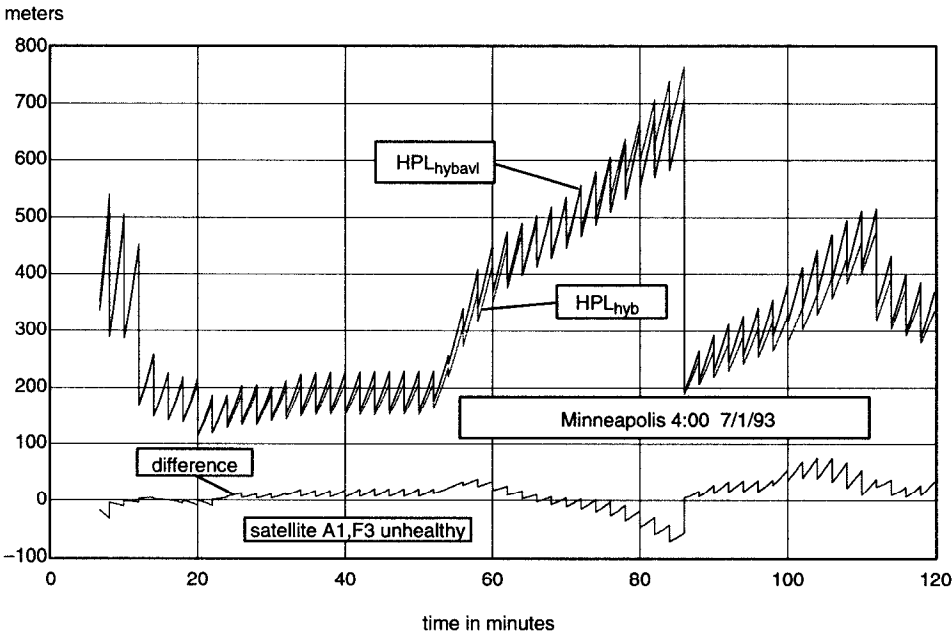


Fig. 5—Simplified Model with Altitude Aiding

allowed 1 h of calibration. Credit is given for any 2 min interval with HPL_{hybavl} below the alert limit.

The results without pressure altitude aiding are given in Tables 6 and 7. When using the lower envelope and a 2 deg mask angle, the results are as shown in Table 6. For the remaining combinations, the weighted averages are as shown in Table 7. The impact of calibrated pressure altitude aiding in

Table 6—GPS/Inertial Fault Detection Availability for Lower Envelope and 2 deg Mask Angle

Constellation	Alert Limits (nmi)			
	0.3	0.6	1.0	2.0
24	0.99992	1.00000	1.00000	1.00000
23	0.99809	0.99993	1.00000	1.00000
22	0.99481	0.99966	1.00000	1.00000
21	0.96900	0.99535	0.99920	0.99998
Weighted	0.99876	0.99990	0.99999	1.00000

Table 7—GPS/Inertial Fault Detection Availability

Mask, Type	0.3 nmi	0.6 nmi	1 nmi	2 nmi
2 deg, upper	0.99729	0.99981	0.99998	1.00000
5 deg, lower	0.99524	0.99945	0.99995	1.00000
5 deg, upper	0.99015	0.99898	0.99989	1.00000

combination with coasting is presented in Table 8. The results for baro-corrected pressure altitude aiding in combination with coasting are presented in Table 9. $\sigma_{altb} = 50$ m, as recommended in FAA TSO-C129.

The following augmentations are included to provide trade-off reference data.

Basic GPS. This is the availability averaged over the constellations in Table 10 with no augmentation.

Calibrated pressure altitude. Calibration of the pressure altitude bias occurs only when the vertical protection level VIL satisfies $VIL/3 < \sigma_{altb}$, where σ_{altb} is the current predicted pressure altitude bias error 1σ . The time propagation of σ_{altb} adheres to the requirements in FAA TSO-C129. An integrity-based calibration of this type is currently mandated by the RTCA SC-159. Calibrated pressure altitude availability results are presented in Table 11.

Exact and fault-free altitude. The altitude is known without noise or faults and can therefore be eliminated in the RAIM algorithm. This provides an upper availability bound for any altitude-aiding augmentation. The exact and fault-free availability results are presented in Table 12.

Table 8—GPS/Inertial Fault Detection Availability with Pressure Altitude Aiding

Mask, Type	0.3 nmi	0.6 nmi	1 nmi	2 nmi
2 deg, lower	0.99966	0.99998	1.00000	1.00000
2 deg, upper	0.99896	0.99997	1.00000	1.00000
5 deg, lower	0.99831	0.99989	0.99998	1.00000
5 deg, upper	0.99557	0.99982	0.99998	1.00000

Table 9—GPS/Inertial Fault Detection Availability with Baro-Corrected Altitude

Mask, Type	0.3 nmi	0.6 nmi	1 nmi	2 nmi
2 deg, lower	0.99992	1.00000	1.00000	1.00000
2 deg, upper	0.99980	1.00000	1.00000	1.00000
5 deg, lower	0.99962	0.99997	1.00000	1.00000
5 deg, upper	0.99910	0.99995	0.99999	1.00000

Table 10—Fault Detection Availability for Basic GPS

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.98849	0.99932	0.99978	0.99992	0.99996
5 deg	0.96686	0.99702	0.99882	0.99954	0.99972

Table 11—Fault Detection Availability for GPS and Calibrated Pressure Altitude

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.99416	0.99990	0.99998	0.99999	1.00000
5 deg	0.97968	0.99940	0.99986	0.99996	0.99998

Table 12—Fault Detection Availability for GPS and Exact Fault-Free Altitude

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.99928	0.99999	1.00000	1.00000	1.00000
5 deg	0.99672	0.99987	0.99996	0.99999	0.99999

Calibrated atomic clock. Calibration of the atomic clock phase occurs only when the time protection level TPL satisfies $TPL/3 < \sigma_{\text{phase}}$, where σ_{phase} is the current predicted clock phase error 1σ . The time propagation of σ_{phase} corresponds to a 10^{-11} frequency stability. The calibrated clock results are presented in Table 13.

Exact and fault-free clock. The clock phase is known without noise or faults and can therefore be eliminated in the RAIM algorithm. This provides an upper availability bound for any clock-aiding augmentation. The exact and fault-free clock results are presented in Table 14.

Pressure altitude aiding and inertial HPL coasting. A reference solution that has been assured by pressure altitude-aided RAIM is propagated based on uncalibrated inertial information. The HPL is propagated through the integrity holes based on a predetermined Schuler-based drift profile that reflects the 2 nmi/h performance. The results are presented in Table 15.

TESTING AND CERTIFICATION

The following off-line tests, in combination with formal verification of simulation model versus actual system, are proposed for the hybrid integrity approach presented in this paper.

- 1) Verify the missed detection probability by simulating 5000 failures (see RTCA DO-208). If no assumptions about the dynamics of the failure have been used in the derivation of the horizontal protection level or in the Kalman filters, as was the case for the algorithm presented, it is sufficient to use ramp (linear) failures. If the calculation of the protection level is based on the assumption that the failure growth is a ramp, other types of failure growth (such as parabolas) must be included in the test. The missed detection probability for the presented algorithm has been verified in 1000 ramp tests.
- 2) Verify the false detection rate by lowering the internal detection threshold, and perform 5000 runs to ensure that the false detection rate corresponds

Table 13—Fault Detection Availability for GPS and Calibrated Atomic Clock

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.99866	0.99996	0.99999	1.00000	1.00000
5 deg	0.99306	0.99976	0.99992	0.99998	0.99999

Table 14—Fault Detection Availability for GPS and Exact and Fault-Free Clock

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.99965	0.99999	1.00000	1.00000	1.00000
5 deg	0.99771	0.99988	0.99996	0.99999	0.99999

Table 15—Fault Detection Availability for Pressure Altitude and Inertial HPL Coasting

Mask	0.3 nmi	1 nmi	2 nmi	5 nmi	10 nmi
2 deg	0.99416	0.99990	0.99998	1.00000	1.00000
5 deg	0.97968	0.99940	0.99986	0.99998	1.00000

to the predicted value based on this modified threshold and assumptions about the simulated SA noise. This was done for the presented algorithm, and correct prediction of the false detection rate was verified using 33.3 m (1σ) white noise SA. With the 2-state SA dither model, the false detection rate was about 8 times too high. This indicates that the recommended test model does not meet the FAA/RTCA SC-159 assumption that SA samples taken 2 min apart can be considered uncorrelated [1]. However, the RTCA assumption that real SA samples taken 2 min apart are uncorrelated is used throughout this paper.

- 3) Show by analysis and simulation that the hybrid protection level (lower envelope) will always be better than the normal parity space-based RAIM algorithm. This was done by demonstrating that the hybrid protection level is obtained by providing the snapshot-based separation and positioning variances as the measurement variances used by the simplified 6-state covariance propagators and Kalman filters. Any Kalman filter with calibration capability will reduce the variance of the white noise measurement variance.
- 4) Use the parity space-based RAIM algorithm to identify integrity holes. Record the location, time, and duration of these holes. Demonstrate the claimed availability by using either a high-fidelity model of the complete hybrid bank or a verified simplified model to show that the outage time accumulated over all holes corresponds to the claimed availability.
- 5) Demonstrate by simulation that the impact of airplane dynamics such as holding patterns is properly incorporated when calculating the horizontal protection level.

It is also recommended that the inertial performance be constantly monitored based on GPS for a system in service, and that any observed significant statistical deviation from the model used for the integrity calculations be flagged as a fault.

CONCLUSIONS

It has been demonstrated that inertial effects such as coasting can be used to enhance the integrity provided by receiver autonomous integrity monitoring (RAIM). Primary-means detection capability is achievable for the en route and oceanic phases of flight. By using an accurate z-accelerometer or altitude aiding, primary-means detection capability can most likely be achieved in the terminal phase as well. Depending on the mask angle used and the capacity of the embedded CPU, an availability of failure detection in the range 0.995–0.9996 for 0.3 nmi and 0.9998–0.99998 for 0.6 nmi alert limits is expected when calibrated pressure altitude aiding is used. The availability is 0.9991–0.99992 for 0.3 nmi and 0.99995–1.00000 for 0.6 nmi alert limits when baro-corrected altitude is used. The 0.99999 availability proposed for sole-means GPS nonprecision approach (0.3 nmi) is not readily obtainable. These results assume that SA is dithered and that the 2 min SA decorrelation time is valid. Similar results can be obtained with SA biases present, but if there are slower-frequency components not meeting the 2 min decorrelation time or the constant bias characteristics, more investigations and further information about SA are needed. Primary-means capability also requires exclusion. Although the bank

of Kalman filters and covariance propagators presented can be extended to perform exclusion for favorable geometries, a rigorous investigation of exclusion availability is not yet available.

APPENDIX A INERTIAL MODELING

The inertial error model used is summarized below. All errors are referenced to the local vertical coordinate frame L.

$$\dot{\boldsymbol{\psi}}^L = \hat{\mathbf{C}}_B^L \Delta \boldsymbol{\omega}_{IB}^B + \boldsymbol{\psi}^L \times (\hat{\boldsymbol{\omega}}_{IE}^L + \hat{\boldsymbol{\omega}}_{EL}^L)$$

$$\Delta \dot{\mathbf{v}}^L = \boldsymbol{\psi}^L \times \hat{\mathbf{C}}_B^L \mathbf{a}_{\text{raw}} + \hat{\mathbf{C}}_B^L \Delta \mathbf{a}_{IB}^B + (2 \hat{\boldsymbol{\omega}}_{IE}^L + \hat{\boldsymbol{\omega}}_{EL}^L) \times \Delta \mathbf{v}^L \quad (\text{A-1})$$

$$+ \begin{bmatrix} -g(0)/R^{(1)} & 0 & 0 \\ 0 & -g(0)/R^{(1)} & 0 \\ 0 & 0 & 2g(0)/R^{(2)} \end{bmatrix} \Delta \mathbf{r}^L$$

$$+ \Delta \mathbf{g}^L \quad (\text{A-2})$$

$$\Delta \dot{\mathbf{r}}^L = \Delta \mathbf{v}^L + \hat{\boldsymbol{\omega}}_{EL}^L \times \Delta \mathbf{r}^L \quad (\text{A-3})$$

where

$\Delta \boldsymbol{\omega}_{IB}^B$ is $\Delta \boldsymbol{\omega}_0^B$ + scale factor and misalignment errors.

$\Delta \mathbf{a}_{IB}^B$ is $\Delta \mathbf{a}_0^B$ + scale factor and misalignment errors.

$\boldsymbol{\psi}^L$ is attitude error in local vertical frame L.

$\Delta \dot{\mathbf{v}}^L$ is velocity error in L frame.

$\Delta \dot{\mathbf{r}}^L$ is position error in L frame.

$\hat{\boldsymbol{\omega}}_{IE}^L$ is estimated rotation of earth-fixed frame E versus inertial frame I transformed to L frame.

$\hat{\boldsymbol{\omega}}_{EL}^L$ is estimated rotation of local vertical frame L versus earth-fixed frame E transformed to L frame.

$\Delta \boldsymbol{\omega}_0^B$ is gyro biases modeled as first-order Markov processes with a 100 h time constant.

$\Delta \mathbf{a}_0^B$ is accelerometer biases modeled as first-order Markov processes (100 h).

$\mathbf{a}_{\text{raw}}$ is measured raw acceleration in body frame B.

$\hat{\mathbf{C}}_B^L$ is transformation matrix from body frame B to local vertical frame L.

$\Delta \mathbf{g}^L$ is gravity deflections and anomaly ($v_x g$, $v_y g$, Δg_z).

R is earth radius.

$g(0)$ is gravity at zero height.

In addition, since the vertical channel needs pressure altitude support for stability, the pressure altitude bias Δaltb is included. This bias is modeled as a first-order Markov process with a 10,000 s time constant. Since pseudorange is measured, the receiver clock also needs to be included. The two states used are the clock phase error Δrtc , expressed as a distance, and the clock frequency error Δvc , expressed as a velocity. The terms that affect integrity are marked in the error model: “(1)” indicates the two terms that are responsible for the Schuler feedback; “(2)” indicates the term responsible for the vertical gravity feedback. The error state vector used in the basic hybrid filters has the following 18 states:

$$\Delta \mathbf{x}^T = (\psi^L, \Delta \mathbf{v}^L, \Delta \mathbf{r}^L, \Delta \text{rtc}, \Delta \text{vfc}, \Delta \omega_0^B, \Delta \mathbf{a}_0^B, \Delta \text{altb}) \quad (\text{A-4})$$

The coasting capability is provided by the continuous estimation of these error states. If SA biases are considered, this state vector can be extended with up to 12 additional states $\Delta \rho_{\text{SA}}$ that represent the bias part of SA, yielding a total of 30 states.

APPENDIX B

THE HYBRID INTEGRITY KALMAN FILTERS AND COVARIANCE PROPAGATORS

The inertial model is linearized, and the time propagation of the error states of the full filter ($n = 0$) and subfilters ($n = 1, N$) can be expressed in time-discrete form ($T = 2$ minutes) as

$$\Delta \mathbf{x}_n(k+1) = \Phi(k) \Delta \mathbf{x}_n(k) \quad (\text{B-1})$$

The error $\delta \mathbf{x}_n(k)$ in the propagated estimate versus the actual inertial offset $\Delta \mathbf{x}(k)$ ($\delta \mathbf{x}_n(k) = \Delta \mathbf{x}_n(k) - \Delta \mathbf{x}(k)$) is propagated as

$$\delta \mathbf{x}_n(k+1) = \Phi(k) \delta \mathbf{x}_n(k) + \mathbf{v}(k) \quad (\text{B-2})$$

Here $\mathbf{v}(k)$ is the process noise, and $\mathbf{Q}(k) = E[\mathbf{v}(k) \mathbf{v}(k)^T]$. The full filter and the subfilter covariances are propagated and updated according to the standard Kalman formulation.

The covariance time propagation is

$$\mathbf{P}_n(k+1) = \Phi(k) \mathbf{P}_n(k) \Phi(k)^T + \mathbf{Q}(k) \quad (\text{B-3})$$

For each GPS measurement except the n^{th} , do the following:

$$\mathbf{g}_n = \mathbf{P}_n(k) \mathbf{h} / (\mathbf{h}^T \mathbf{P}_n(k) \mathbf{h} + r) \quad (\text{B-4})$$

$$\mathbf{P}_n(k) = \mathbf{P}_n(k) - \mathbf{g}_n \mathbf{h}^T \mathbf{P}_n(k) \quad (\text{B-5})$$

$\mathbf{h}$ defines the error states that are involved in the GPS measurement. r is the measurement noise variance, and $r = E[w^2] = \sigma_{\text{pr}}^2$.

$$\mathbf{h}^T = (0, 0, -\mathbf{los}^L, 1, 0, 0, 0, 0) \quad (\text{B-6})$$

$\mathbf{los}^L$ is the line-of-sight vector to the satellite. The covariance propagator keeps track of the statistics of the separation $\mathbf{d}\mathbf{x}_n(k) = \Delta \mathbf{x}_0(k) - \Delta \mathbf{x}_n(k)$ by forming a state vector

$$\delta \mathbf{X}(k) = \begin{bmatrix} \delta \mathbf{x}_0(k) \\ \mathbf{d}\mathbf{x}_n(k) \end{bmatrix} \quad (\text{B-7})$$

This vector is time propagated as

$$\begin{bmatrix} \delta \mathbf{x}_0(k) \\ \mathbf{d}\mathbf{x}_n(k) \end{bmatrix} = \begin{bmatrix} \Phi(k) & \mathbf{0} \\ \mathbf{0} & \Phi(k) \end{bmatrix} \begin{bmatrix} \delta \mathbf{x}_0(k) \\ \mathbf{d}\mathbf{x}_0(k) \end{bmatrix} + \begin{bmatrix} \mathbf{v}(k) \\ \mathbf{0} \end{bmatrix} \quad (\text{B-8})$$

resulting in a covariance propagation

$$\mathbf{P}_n^{\text{dual}}(k+1) = \begin{bmatrix} \Phi(k) & \mathbf{0} \\ \mathbf{0} & \Phi(k) \end{bmatrix} \mathbf{P}_n^{\text{dual}}(k) \begin{bmatrix} \Phi(k)^T & \mathbf{0} \\ \mathbf{0} & \Phi(k)^T \end{bmatrix} + \begin{bmatrix} \mathbf{Q}(k) & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{bmatrix} \quad (\text{B-9})$$

For each satellite measurement except the n^{th} , the covariance vector is updated based on

$$\begin{bmatrix} \delta \mathbf{x}_0(k) \\ \mathbf{d} \mathbf{x}_n(k) \end{bmatrix} = \Phi \begin{bmatrix} \delta \mathbf{x}_0(k) \\ \mathbf{d} \mathbf{x}_n(k) \end{bmatrix} + \Gamma \mathbf{w} \quad (\text{B-10})$$

with

$$\Phi = \begin{bmatrix} \mathbf{I} - \mathbf{g}_0 \mathbf{h}^T & \mathbf{0} \\ (\mathbf{g}_n - \mathbf{g}_0) \mathbf{h}^T & \mathbf{I} - \mathbf{g}_n \mathbf{h}^T \end{bmatrix} \quad \text{and} \quad \Gamma = \begin{bmatrix} \mathbf{g}_0 \\ \mathbf{g}_0 - \mathbf{g}_n \end{bmatrix}$$

The result is a covariance update yielding

$$\mathbf{P}_n^{\text{dual}}(k) = \Phi \mathbf{P}_n^{\text{dual}}(k) \Phi^T + \Gamma \mathbf{r} \Gamma^T \quad (\text{B-11})$$

$\mathbf{P}_n$ and $\mathbf{d} \mathbf{P}_n$ contain all information about the 2-D elliptical distributions in the horizontal plane. Special rules apply to the initialization and release of the subfilters and propagators to ensure that a satellite failure in the excluded satellite can never affect the subfilter associated with this satellite, and that at least one subfilter is kept that is always unaffected by any satellite failure. When a new satellite is providing measurements, a new subfilter is initialized with the full filter solution, which is unaffected by any failure in this new satellite. When the pseudorange measurements from a satellite are lost and the corresponding subfilter is to be released, special logic will ensure that the impact of a failure in this satellite is not propagated in the remaining subfilters. This is done by delaying the release of any subfilter until the HDOPs of all subfilters are below a predetermined threshold for at least 5 min.

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